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PROFESSIONAL SUMMARY

Data Scientist & Engineer at Tata 1mg, building end-to-end ML systems and GenAI applications across Homepage & Search sciences and healthcare intelligence. Experienced in taking models from research to production, from knowledge distillation and VLM-based pipelines to search re-ranking and A/B experimentation at scale. Skilled in Python, prompt engineering, and real-time event data, with a strong foundation in backend systems that most DS profiles lack. Active tech community contributor, speaker at Microsoft India, IEEE WIE, and EMPI Business School, and mentor at India AI Impact Summit 2026.

WORK EXPERIENCE

Data Scientist-1 | Tata 1mg

September 2025 - Current

(Homepage & Search Sciences, GenAI, Healthcare Intelligence)

- Engineered external report digitisation pipeline using an in-house VLM to extract medical parameters from non-1mg reports, scaling to ~7 lakh reports digitised; mapped outputs to internal catalog via BM25 + Milvus vector embeddings with semaphore-based async execution for high-throughput processing.
- Executed knowledge distillation (teacher-student) from GPT-o3 → GPT-4.1-mini on ~1,000 curated patient health insights; generated teacher rationales, fine-tuned student outputs, and built an evaluation harness to retain explanation quality while cutting inference latency and cost.
- Built benchmarking framework evaluating Gemini 2.5 Flash, GPT-o3, GPT-4o, GPT-5, and Qwen 3.5 across MMLU, PubMedQA, and MedQA; analysis directly drove production model selection decisions.
- Currently leading Homepage and Search sciences; conducting deep-dive EDA in Databricks and designing A/B experiments via Mixpanel/GA to improve CTR and Search-to-Cart conversion.

Backend Engineer | Tata 1mg

January 2024 - September 2025

- Designed and built the health assistant chatbot backend ground up, owning system architecture across 6 chat flows (session management, history retrieval, feedback, issue reporting, orders, and PDP); drove 2x growth in daily footfall (1,000 → 2,000/day) post-launch while optimising API costs and implementing robust error handling
- **Restructured legacy services** to integrate OpenAI and Gemini APIs, enabling dynamic prompt and feature configuration with all prompts stored and managed via the database for extensibility and controlled rollout.
- Designed and implemented a **four-stage AI-driven doctor consultation flow**, enabling structured data collection, report explanations, and reasoning over **comorbidities** to generate comprehensive patient health overviews and support personalized digital health insights.
- **Version upgraded** the torpedo framework within the AI repository by upgrading core dependencies (pydantic, beanie, pymongo, genai), refactoring schema validation and async handling to achieve faster response times and reduced runtime errors.
- Built patient-facing notification and recommendation APIs, **automating report delivery** and enabling lifestyle, diet, and supplement recommendations through rule-based and scheduled backend workflows.
- Co-developed an **On-call Monitoring Dashboard** to automate incident handling for CX-reported issues, replacing manual Kubernetes workflows with a centralized, auditable interface and significantly reducing mean time to recovery (MTTR); **mentored an intern** who was successfully converted to FTE.

Software Development Intern | Essentia Dev

June 2023 - August 2023

- Designed and developed intuitive **profile and dashboard interfaces** (3-4 pages) for a travel booking application, improving overall usability and accessibility.
- Implemented multi-tenancy by **restructuring database tables**, reducing application replication costs by 75%.
- Authored a blog detailing the **multi-tenancy implementation**, as the only intern among 20+ peers to have their work published on the company's official site.

Technical Intern | Indraprastha Gas Limited

August 2022 - September 2022

- Built digitization of gatepass procedures, creating web portal with facility of Java web forms to monitor the ingress and egress of office materials like office authorized laptops, Document files etc..
- Utilized iText jar files, generated invoices for transferred items, subject to approval by security and IT authorities and digitized 1000+ such forms.

Blockchain Research Intern | Gail(India) Limited

July 2021 - August 2021

- Engineered a blockchain ledger system for the transparent transportation of gasoline, minimising corruption by eliminating intermediaries.
- Proficiently composed, validated, and deployed 4 Solidity smart contracts, leveraging rinkeby ether to draw a digital legal contract that ensure payment while transportation as a form of a ledger and led a thorough Proof of Concept, and presented my work to a team of professionals at GAIL.

SKILLS

- **Languages:** Python, SQL, Bash
- **Product Data Science:** A/B Experimentation, Personalization, Relevance Modeling, Metric Design, Offline & Online Evaluation
- **ML / AI:** Ranking & Recommendation Systems, LLM Fine-Tuning, Prompt Engineering, LLM Evaluation & Benchmarking, Knowledge Distillation
- **GenAI & Retrieval:** RAG Architectures, OpenAI APIs, Gemini, Milvus, FAISS, BM25, Assistant-Thread APIs
- **Backend & Systems:** API Design, Async Systems, Distributed Services, System Design
- **Data & Infra:** Databricks, Airflow, Mixpanel, Google Analytics
- **Databases:** PostgreSQL, MongoDB
- **Tools:** Django, Sanic, Redash, Sentry, GitHub

PROJECTS

Brain Tumour Detection - Research Work (ongoing)

- Collaborating with Dr. Chakresh Jain (JIIT) to build a YOLOv7-based deep learning system for brain tumour detection and stage classification from medical imaging; developing end-to-end pipeline covering preprocessing, training, evaluation, and inference optimisation.

Prescription QnA System (Healthcare NLP / GenAI)

- Built a structured extraction pipeline to identify medical entities (drug names, dosages, frequencies) from prescriptions and integrated LLMs to generate context-aware responses with hallucination minimisation via prompt constraints.
- Implemented retrieval and grounding mechanisms ensuring answers were strictly based on extracted prescription content, improving reliability in a high-stakes healthcare context.

SPEAKING & COMMUNITY

- **Speaker — Microsoft India Sovereign, Azure Developer Community**
 - Delivered talk "All About AI" on LLMs and AI in healthcare to IIMT University students
- **Panelist — IEEE WIE JSSATEN x WIEJSSUN, Women's Day Panel**
 - Spoke on collaborative problem solving and women's role in tech; shared applied AI work from Tata 1mg
- **Guest Lecturer — EMPI Business School**
 - Delivered session on Market Analytics & Customer Insights; designed hands-on data-driven campaign activity for students
- **Mentor — IDEATRON, India AI Impact Summit 2026 (Netweb Initiative)**
 - Mentored teams building AI solutions across healthcare, agriculture, and education
- **Judge — SnapAR Hackathon (Snap Inc.) by BharatXR**
 - Evaluated AR projects built by college students

EDUCATION

Jaypee Institute Of Information Technology - Btech (CSE)	2020-24	CGPA: 7.5
Ahlcon Public School (12th Board)	2019-2020	94%
Ahlcon Public School (10th Board)	2018-2019	93.6%

CERTIFICATIONS

1. NPTEL - Reinforcement Learning
2. NPTEL - Affective Computing
3. Django: python web development unleashed

BLOG

1. [Multitenancy for Dummies](#)
2. [Demystifying Reinforcement Learning](#)
3. [Multi-arm Bandit: Exploration vs Exploitation](#)